

FIELD OF INVENTION

[001] The field of invention for a **Smart Grid-Based Prepaid Metering System** typically falls within the domain of **smart grid technology** and **advanced metering infrastructure (AMI)**. Here are the key areas of invention and innovation in this field:

1. **Smart Grid Technology:** This includes the integration of digital communication and monitoring systems into the electrical grid to enhance efficiency, reliability, and management. Inventions in this area may focus on improving grid stability, real-time data analysis, and communication protocols.
2. **Prepaid Metering Systems:** These systems allow users to pay for electricity in advance, similar to a prepaid mobile phone. Innovations here might involve more sophisticated user interfaces, enhanced payment options, or better integration with billing systems.
3. **Advanced Metering Infrastructure (AMI):** This encompasses the technologies and systems used to collect, store, and analyze data from utility meters. Inventions could include more accurate sensors, better data encryption methods, or improved data transmission techniques.

[002] In summary, the field of invention for a smart grid-based prepaid metering system involves a blend of smart grid advancements, metering technology, communication innovations, and user-centered design improvements. Each of these areas offers opportunities for new solutions and improvements to enhance the efficiency and effectiveness of modern electrical grids.

BACKGROUND OF THE INVENTION AND PRIOR ART (Separate page)

[003] A **Smart Grid-Based Prepaid Metering System** represents a significant advancement in the management of electrical energy distribution and consumption. This technology combines the principles of smart grid systems with prepaid metering to offer enhanced control and efficiency for both utilities and consumers. The background of this invention can be understood in several contexts:

1. **Evolution of the Electrical Grid:**
 - Traditional electrical grids are designed for unidirectional power flow from centralized power plants to consumers. This model often lacks

real-time data, making it difficult to manage demand and supply effectively.

- The smart grid paradigm introduces two-way communication between utilities and consumers, allowing for real-time monitoring, control, and optimization of the grid.

2. Need for Advanced Metering:

- Conventional meters record consumption but do not provide detailed data or real-time feedback. Users receive bills after the fact, which can lead to billing disputes and inefficient energy use.
- Advanced Metering Infrastructure (AMI) addresses these limitations by providing real-time data on energy usage, enabling more accurate billing and better energy management.

3. Prepaid Metering Systems:

- Prepaid metering allows users to pay for energy in advance, similar to prepaid mobile phone plans. This system helps users manage their energy budgets more effectively and reduces the risk of arrears.
- Prepaid meters can offer various payment options, including online payments, mobile applications, and physical vouchers.

4. Integration of Smart Grid and Prepaid Systems:

- Combining smart grid technology with prepaid metering systems enhances the capabilities of both. It enables real-time monitoring of energy usage, better demand management, and dynamic pricing models.
- Smart grids with prepaid metering systems can provide users with detailed consumption data, alerts, and notifications, improving overall energy management and efficiency.

OBJECT OF THE INVENTION

[004] The **object of the invention** for a **Smart Grid-Based Prepaid Metering System** is to develop a comprehensive solution that integrates advanced metering technology with the capabilities of modern smart grids to enhance the management, efficiency, and user experience of energy consumption. Specifically, the objectives include:

1. Real-Time Monitoring and Management:

- **Objective:** To provide real-time monitoring of energy consumption and grid conditions.
- **Benefit:** Allows both utilities and consumers to have up-to-date information on energy usage, which facilitates better management and decision-making.

2. Enhanced Prepaid Billing:

- **Objective:** To enable a flexible and user-friendly prepaid billing system that integrates seamlessly with smart grid technology.
- **Benefit:** Provides consumers with control over their energy expenditures, helps manage budgets, and reduces the risk of arrears.

3. Improved Data Accuracy and Security:

- **Objective:** To ensure accurate measurement of energy consumption and secure transmission of data between the meter and the utility.
- **Benefit:** Minimizes billing errors, enhances trust in the system, and protects sensitive user data from unauthorized access.

4. Dynamic Pricing and Demand Response:

- **Objective:** To support dynamic pricing models and demand response strategies through real-time data and communication.
- **Benefit:** Enables utilities to optimize energy distribution and pricing based on current grid conditions, and helps consumers adjust their usage to take advantage of lower rates or avoid peak pricing.

5. User Engagement and Feedback:

- **Objective:** To offer interactive features and feedback mechanisms that improve user engagement with their energy consumption.

- **Benefit:** Helps consumers understand and manage their energy use more effectively through alerts, usage reports, and consumption trends.

6. **Seamless Integration with Smart Grid Infrastructure:**

- **Objective:** To integrate prepaid metering with existing smart grid infrastructure for streamlined operation and management.
- **Benefit:** Ensures compatibility with other smart grid components, such as grid sensors and communication networks, enhancing overall grid efficiency and reliability.

7. **Scalability and Flexibility:**

- **Objective:** To design a system that is scalable and adaptable to various grid sizes and configurations.
- **Benefit:** Accommodates future growth and technological advancements, making the system suitable for a wide range of applications from small residential to large industrial environments.

8. **Cost Efficiency:**

- **Objective:** To develop a cost-effective solution that balances initial investment with long-term savings and benefits.
- **Benefit:** Provides an economically viable option for utilities and consumers, ensuring that the deployment of smart prepaid metering systems is feasible and beneficial.

9. **Regulatory Compliance:**

- **Objective:** To ensure that the system meets regulatory standards and requirements for metering, billing, and data security.
- **Benefit:** Facilitates compliance with industry regulations and standards, reducing the risk of legal and operational issues.

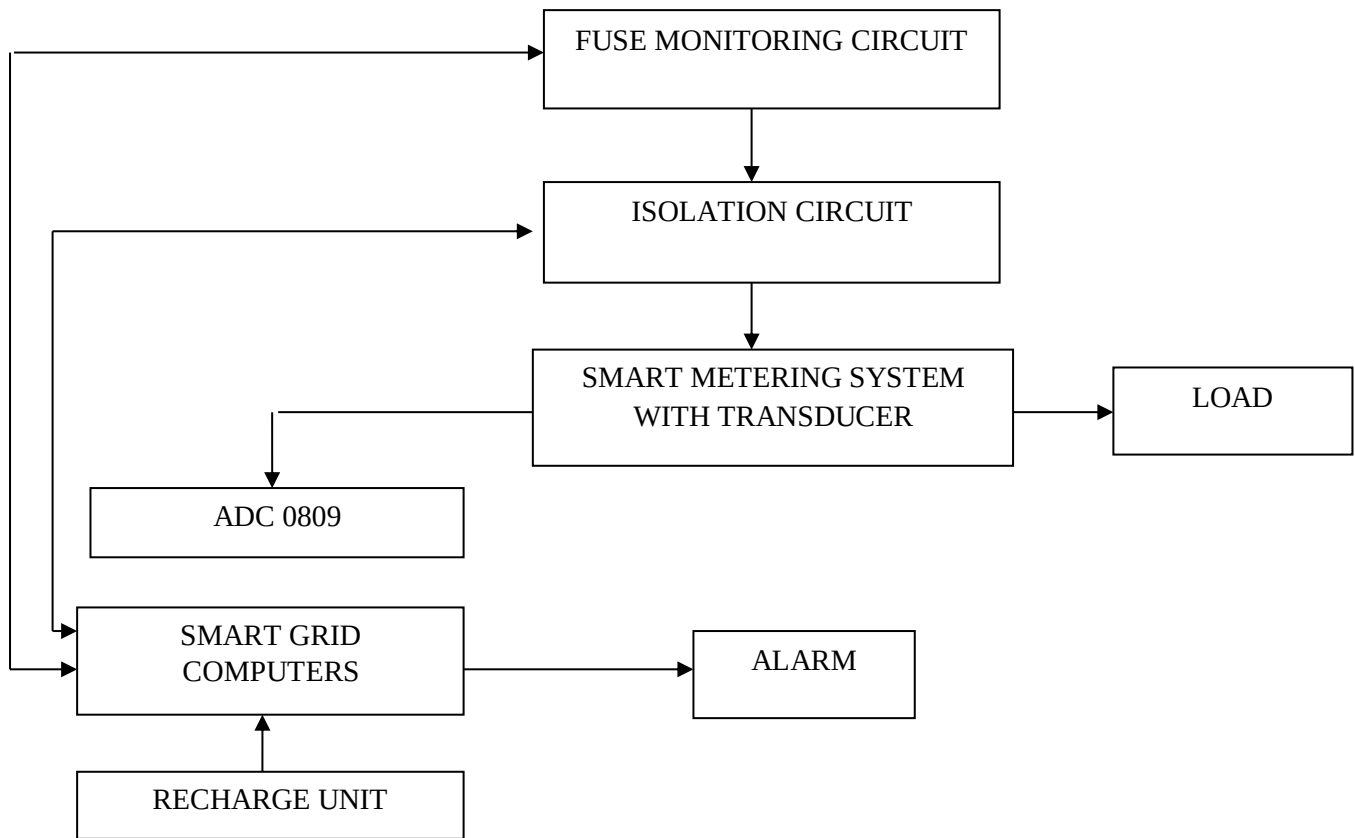
10. **Integration with Renewable Energy Sources:**

- **Objective:** To support the integration of renewable energy sources into the grid, allowing for effective management of variable energy inputs.
- **Benefit:** Enhances the sustainability of energy systems and supports the transition to cleaner energy sources.

SUMMARY OF THE INVENTION: (Separate page)

[005] The **Smart Grid-Based Prepaid Metering System** is an advanced solution that integrates smart grid technology with prepaid metering to offer a more efficient, user-friendly, and flexible approach to managing electricity consumption and billing. This invention aims to enhance the capabilities of both smart grids and prepaid metering systems, providing significant benefits to both utility providers and consumers. The system is designed to address the limitations of traditional metering and billing methods by incorporating real-time data, dynamic pricing, and advanced communication features. The Smart Grid-Based Prepaid Metering System represents a significant advancement in energy management technology. By combining real-time data monitoring, prepaid billing, dynamic pricing, and seamless smart grid integration, the system offers a comprehensive solution that improves efficiency, user engagement, and overall grid performance. It addresses the shortcomings of traditional metering systems and provides a flexible, scalable, and cost-effective approach to modern energy management.

BRIEF DESCRIPTION OF THE DRAWINGS: (Separate page)



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DETAILED DESCRIPTION

SMART METERS

Smart meters are a new kind of energy meter. Smart meters are a replacement for your existing meters and send electronic meter readings to your energy supplier automatically.

Smart meter works by communicating directly with your energy supplier, so the company will always have an accurate meter reading and there's no need for you to take a meter reading yourself.

TRANSDUCER

The Power system deals with high voltage, high current and different electrical parameters. To measure or to monitor these parameters some circuits/hardware are essential and this hardware otherwise called as transducers. The transducers can handle different input parameters like voltage, power factor, frequency, current, etc. The standard output specified is DC voltage of range 0-5v or DC voltage of range 1 to 5v (or) DC current of range 0-20mA (or) DC current of range 4 to 20mA.

ANALOG TO DIGITAL CONVERTER

An analog-to-digital converter (abbreviated ADC, A/D or A to D) is a device that converts a continuous physical quantity (usually voltage) to a digital number that represents the quantity's amplitude.

The functional components involved in analog to digital conversion circuit: ADC0809, 555 Timer, 74HC240 and 74HC245. The interfacing circuit consists of two buffers IC's 74HC244, which is similar to 74HC240, but non-inverting.

ADC 0809

The ADC0809 peripheral device is a data acquisition interface universally known as an Analog to Digital Converter. This component is a CMOS device with an 8-bit analog to digital converter, 8 channel multiplexed & microprocessor compatible

control logic. This device is widely used owing to its flexibility in programming and configuring the device for various applications in real time.

POWER THEFT BOARD

Power theft detection board is designed with two consumers, two theft point and one source. All the connectors from various check meters are given to power theft board. The input terminal supplies power to other parts of hardware through a transformer.

RELAY AND ISOLATION DRIVE CIRCUIT:

The main components are

- Transistor
- Resistors (2.2 K Ω , 330 Ω)
- Attractive Relay
- Lamp, LED
- Diode, Capacitor (10 μ f)

Operation of Relay and Drive circuit is

The command from the PC is connected to the relay drive circuit. The relay drive circuit consists of medium power transistor wired in the switching mode. When an input voltage of sufficient magnitude is applied to the base of the transistor, the transistor goes into saturation, turning on the relay connected at its collector terminal. A diode is used as a free wheeling element to prevent the induced voltage in the relay coil from damping the transistor when relay driver transistor is cut-off.

11 RECHARGE UNIT

A standard 10 pin edge connector is used to connect recharge card. Edge connector consists of 10 binary inputs, pin 1,10 is connected to GND (0V) and pin 4 is connected with 5Volts. Two-digit binary code through pins 2 and 3 is extended to interface card. Six-digit validity code is extended to interface card through pins 5 to 10. When the card is inserted two-digit digital input is extended

to computer for indicate the type of the card to assign the amount of recharge. The digit 00 represent the minimum amount Rs.0.10, digit 01 represent the recharge amount Rs.0.20 , The recharge amount is programmable and can be altered. The six digit codes digit is extended to computer through the recharge card to identify the card as valid. The six-digit code 111000 is assigned for recharge card 1 and code 110000 is assigned for recharge card 2.

CLAIM:

The Smart Grid Based Prepaid Metering System uses cutting-edge communication module consisting of Wi-Fi, cellular, and LoRaWAN networks.

The Smart Grid Based Prepaid Metering System the prepaid credit management system further comprises an authentication module to verify prepaid transactions and prevent fraud.

The Smart Grid Based Prepaid Metering System the central server includes a data analytics module configured to analyze consumption patterns and predict future usage.

The Smart Grid Based Prepaid Metering System provides notifications to users via a mobile application when the prepaid credits fall below a predetermined threshold.

The Smart Grid Based Prepaid Metering System includes a mobile application for users to view and manage their prepaid balance and energy consumption.

The Smart Grid Based Prepaid Metering System comprising a backup power supply unit integrated with the smart meter to ensure continued operation during power outages.

The Smart Grid Based Prepaid Metering System includes a local display for users to view real-time consumption and credit balance.

The Smart Grid Based Prepaid Metering System is configured to send automated notifications to users via SMS or email when their prepaid credit balance is low.

The Smart Grid Based Prepaid Metering System the prepaid credit management unit includes a module for tiered pricing based on consumption levels.

The Smart Grid Based Prepaid Metering System is further configured to support integration with third-party energy management systems for enhanced monitoring and optimization.

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ABSTRACT

The present invention relates to a Smart Grid Based Prepaid Metering System designed to integrate with modern smart grid infrastructures for efficient management of electrical consumption and billing. The system comprises a smart meter equipped with real-time consumption measurement and a communication module for data exchange with a central server. The central server manages prepaid credit balances for multiple smart meters, updates credit information, and monitors energy usage patterns. The prepaid credit management unit within each smart meter handles credit storage, deduction based on consumption, and generates alerts when the balance is low. Users and utility providers can access a user interface, including mobile applications and web portals, to manage credits, view usage data, and receive notifications. The system enhances energy management by providing real-time feedback, preventing over-consumption, and enabling flexible billing options. It supports various communication networks and includes features such as authentication, security, and predictive analytics for optimized energy use and maintenance.

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